

```
inputs = [1.0,2.0]

weights = [0.5,-0.6]
bias = 0.1

output_neuron = inputs[0] * weights[0] + inputs[1] * weights[1] + bias

output_neuron
```

-0.6

```
#Activation function (sigmoid) ->flats the value in the range of 0 and 1
import math
def sigmoid(x_val):
    return 1/(1+math.exp(-x_val))
```

```
sigmoid_output = sigmoid(output_neuron)
sigmoid_output
```

0.35434369377420455

```
def relu(x_val):
    return max(0,x_val)
```

```
relu_output = relu(output_neuron)
relu_output
```

0

```
import numpy as np
def forward_propagation(inputs, weights, bais):
    wx = np.dot(inputs, weights)
    z = wx + bais
    sigmoid_output = sigmoid(z)
    relu_output = relu(z)
    return z, sigmoid_output, relu_output
```

```
forward_propagation(inputs, weights, bias)
```

```
(np.float64(-0.6), 0.35434369377420455, 0)
```